# Symbols used in the README of Pre-gravity6plane

| symbol | definition |
|---|---|
| $x_0..x_7$ | coordinates: hidden, space (1-3), time (4), superluminal time (5-7) |
| $\eta$ | $\mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ |
| $\tau_A$, $\bar{\tau}_A$ | real 8x8 matrices of Pre-gravity6plane.nb, $\bar{\tau}_A = \sigma\tau_A^T\sigma$, $\sigma = \begin{pmatrix} 0 & I_4 \\ I_4 & 0 \end{pmatrix}$ |
| $T^A = (T16^A)[A]$ | $\begin{pmatrix} 0 & \bar{\tau}_A \\ \tau_A & 0 \end{pmatrix}$, real Dirac matrices |
| $S^{AB}$ | $\frac{1}{4}[T^A, T^B]$ |
| $\Gamma$, $P_{L,R}$, $\sigma_{16}$ | $T^0\cdots T^7$; $\frac{1}{2}(I_{16} \mp \Gamma)$; $T^0T^1T^2T^3$ |
| $\psi$, $\bar{\psi}$ | dirac16complex (complex Grassmann 16-spinor), $\bar{\psi} = \psi^\dagger\sigma_{16}$ |
| $\mathcal{L}$, $m$, $\lambda$ | Lagrangian, mass, quartic coupling |
| $K$ | $-i\sigma_{16}T^4$ |
| $\psi_\pm$, $S_\pm$, $\mathcal{L}_D$ | $K\psi_\pm = \pm\psi_\pm$; $S_\pm = \psi_\pm^\dagger(-iT^4)\psi_\pm$; $\mathcal{L}_D[\chi] = \chi^\dagger(-iT^4)(T^A\partial_A - m)\chi$ |
| $M$, $H$ | mass and Hubble parameters of Pre-gravity6plane.nb; $m = -2M$ |
| $z$, $t$, $p$, $q$ | $z = 6Hx_0$, $t = Hx_4$, mode $e^{pz+qt}u$ |
| ${F_A}^a$, ${F_a}^A$, $V_1$, $V_2$ | triality frames and vector forms: $V_1 = \eta(\psi_1^T\sigma{F_A}^a)$, $V_2 = {F_a}^A\psi_2$ |
| $n_k$, $Z$, $M^6$ | $n_k = V_k[\mathrm{Re}\,\psi]$; $Z \in \mathbb{R}^8$; $M^6 = \{Z : n_1\cdot Z = n_2\cdot Z = 0\}$ |
| $f$, $r_s$, $H_\Lambda$ | $1 - r_s/x_0$ (modern), $1 - r_s/x_0 - H_\Lambda^2 x_0^2$ (future) |
| $\gamma^\mu$, $D_\mu$ | ${E_A}^\mu T^A$; $\partial_\mu + \frac{1}{4}{\omega_\mu}^A{}_B T_A T^B$ |
| $\rho_s$, $h_{KS}$, $\alpha$, $\beta$, $p$, $B$ | Kohn-Sham density, $\alpha p + BM\beta$, $-T^4T^0$, $-iT^4$, $-i\,d/d\rho$, lapse $e_4$ |
| $\Omega(T)$ | Kohn-Sham grand potential at temperature $T$ |
| $R_1$ | $x_1 \mapsto -x_1$ |
| $R$, $\kappa$, $T_H$, $\epsilon$ | scalar curvature, surface gravity, $\kappa/2\pi$, Killing energy |